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Numerical modeling and experimental validation of nanosecond laser welding parameters for glass-aluminum bonding

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ABSTRACT

An integrated numerical and experimental investigation of direct laser welding between BOROFLOAT® 33 glass and AA1050 aluminum is presented to address the persistent challenges arising from their substantial thermal and mechanical property mismatches. A three-dimensional COMSOL Multiphysics model was developed to simulate transient heat transfer and thermally induced stress under a wobbling nanosecond laser, incorporating a volumetric heat source for glass and surface heat sources for aluminum. To predict crack initiation and propagation in the brittle glass, a fully coupled thermo-mechanical phase-field model was implemented, enabling the continuous evolution of damage driven by thermal gradients and residual stresses. Simulation results reveal that high laser power and low scanning speed significantly elevate damage-driving energy and promote cracking, whereas moderate energy input confines stress localization. Experimental welding trials validated these predictions, showing close correspondence between simulated damage fields and observed molten morphology, pore formation, and crack behavior. To further optimize process parameters, artificial neural networks (ANNs) were trained on phase-field damage outputs to construct a high-resolution processing map. The resulting map identifies an optimal low-damage window at ~30–50 W with a scanning speed of 3–5 mm/s, yielding crack-free seams and a maximum shear strength of 8.5 MPa—surpassing many reported glass-metal welding benchmarks. This combined computational–experimental framework provides a quantitative strategy for optimizing glass–metal laser welding without interlayers, offering a robust foundation for future optoelectronic, microfluidic, and packaging applications.

1. Introduction

Laser welding is a highly precise joining technique that uses focused laser energy to melt and fuse materials [1–4]. When applied to dissimilar materials such as glass and aluminum, the process presents unique challenges due to their fundamentally different thermo-mechanical properties [5]. The direct laser joining of borosilicate glass (BF33) and aluminium (AA1050) is primarily inhibited by their fundamentally disparate thermal and physical properties. A critical challenge arises from the high thermal conductivity ratio (Al~222 W/(m·K) vs. BF33~1.2 W/(m·K), which creates a massive asymmetry in heat dissipation at the interface, leading to highly localized thermal gradients exceeding 106 K/s during nanosecond pulses [6]. Furthermore, the significant mismatch in the Coefficient of Thermal Expansion (CTE), where AA1050 ($24x10^{-6}$/K) is approximately seven times higher than BF33 (~$3.25x10^{-6}$/K) [7], induces substantial interfacial residual stresses (σ_{res}) during the rapid cooling phase. Following the simplified relationship for thermal stress, $\sigma = E\,\alpha\,\Delta T$, this mismatch directly triggers brittle fracture when the induced tensile stress exceeds the local fracture toughness of the glass. These quantitative contrasts explain the high susceptibility to micro-cracking and provide the physical motivation for the phase-field damage modelling and 'wobbling' stress-relief strategies employed in this study [8]. Glass is brittle and highly sensitive to thermal gradients, whereas aluminum is ductile and exhibits high thermal conductivity. These contrasts make it difficult to achieve strong and reliable joints without inducing thermal distortion, interfacial stress, or cracking. Despite these challenges, glass–metal welding has important applications in electronics, where hermetic sealing is required for sensors, batteries, and optical devices [9–11]. It is also used in lightweight structural components within the automotive and aerospace sectors [12,13], and is increasingly adopted in architectural designs that integrate transparency with mechanical robustness. The primary

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advantage of laser welding in these contexts is its ability to provide localized heating with minimal heat-affected zones (HAZ), enabling controlled melting and high-precision bonding.

To address the challenges associated with glass–metal joining, both direct and indirect welding strategies have been explored [14,15]. Direct welding relies solely on laser-induced melting of the interface, but the large thermal-expansion mismatch often leads to crack formation and reduced bond strength. Indirect methods introduce an intermediate layer—such as micro-arc oxidation (MAO) coatings, metallic interlayers, or surface treatments—to buffer thermal stresses and improve wettability. For example, Min et al. [11] achieved hermetic sealing between ultrathin glass and aluminum using MAO-coated surfaces, while other studies have shown that copper or titanium interlayers can improve stress distribution and bonding stability. However, interlayers introduce added processing steps and potential issues, including contamination, interfacial diffusion, and altered optical or mechanical performance. These limitations motivate continued development of direct, interlayer-free welding approaches.

Recent studies have therefore focused on optimizing direct laser welding parameters—such as power, scanning speed, pulse duration, and spot overlap—to enhance interface quality and suppress crack formation [14–18]. Surface preparation also plays a crucial role; for example, pre-oxidation of aluminum has been shown to enhance the strength and uniformity of glass–metal welding [14]. Advances in ultrafast laser technology have enabled the successful joining of glass to metals, including copper and stainless steel, even without mechanical clamping [16,17]. In glass–glass systems, femtosecond lasers with water-based intermediate layers have achieved high bonding strength while avoiding optical-contact constraints [18]. In addition, to better understand crack initiation mechanisms in brittle materials during laser processing, phase-field fracture modeling has emerged as a powerful tool. This method represents damage evolution through a continuous scalar field, capturing crack nucleation, propagation, and branching in response to stress concentrations and thermal gradients. Coupled thermo-mechanical phase-field formulations have been successfully used to predict thermally induced fractures under various high-temperature manufacturing conditions [19–21]. These models offer valuable insights into how laser parameters affect thermal stress and interfacial integrity.

To overcome the inherent challenges of thermal stress and brittle fracture in the joining of dissimilar brittle materials, recent cutting-edge research has pivoted toward advanced laser beam delivery and interface modification strategies. For instance, recent studies have demonstrated that the application of ultrafast laser oscillation in the welding of yttria-stabilized zirconia (YSZ) ceramics can significantly refine joint microstructure by smoothing the morphology of the ablation zone and replacing sharp triangular notches with "finger-like" structures, resulting in a 90% improvement in bending strength compared to direct fusion welding [22]. Parallel research has emphasized the critical role of metal intermediate layers in stabilizing the molten pool and optimizing heat distribution during laser transmission welding (LTW). Specifically, investigations into the use of Sn film as a metal absorbent for joining polyphenylsulfone resin found that the metallic layer effectively suppressed bubble formation and narrowed temperature fluctuations while promoting the formation of new chemical bonds (eg, Sn–C) that increased joint shear strength by over 25% [23]. These advancements highlight an ongoing scientific shift toward utilizing specific metallic interfaces and dynamic beam control to tailor the thermo-mechanical properties of laser-welded joints, a strategy that is fundamental to the glass-aluminium bonding parameters investigated in this work. In addition, achieving high-strength hermetic seals in glass–metal heterostructures necessitates a delicate equilibrium between metallurgical bond density and processing efficiency. Recent investigations utilizing nanosecond fiber lasers have demonstrated direct welding of aluminum alloys to borosilicate glass via beam oscillation. However, these spatially modulated nanosecond processes often result in localized residual stress and excessive growth of intermetallic compounds, which typically limit shear strengths to 12–15 MPa [24]. On the other hand, while ultra-short-pulse regimes achieve higher joint integrity, their lower scanning speeds generally do not meet high-throughput manufacturing requirements [25]. Consequently, nanosecond processing favors speed but suffers from strength limitations, whereas femtosecond processing excels in bond quality but is less compatible with rapid production. Moreover, as noted in mechanistic studies [26], precise control over the thermal history is critical; it enables atomic diffusion necessary for bonding, yet must be balanced to prevent brittle fracture.

In this work, we combine numerical simulations and experimental validation to investigate glass–aluminum laser welding, with a focus on the relationships between heat distribution, stress evolution, and interfacial cracking. A comprehensive simulation framework, including transient heat transfer, thermo-mechanical coupling, and phase-field fracture, was developed to predict crack risk under various processing conditions. Complementary experiments were conducted using a wobbling-beam nanosecond laser, allowing systematic evaluation of weld morphology, defect formation, and mechanical performance. While laser welding of dissimilar materials has been explored, this study specifically addresses the limitations of conventional approaches by integrating a novel wobbling nanosecond laser technique with a predictive phase-field-ANN framework. Unlike static laser beam irradiation, which induces excessive localized heat accumulation and brittle fracture due to limited thermal diffusion in glass, the high-frequency wobbling technology improves energy uniformity by redistributing heat over a wider interfacial area. This acts as an *in-situ* stress relief mechanism, reducing the thermal gradient ($\Delta T/\Delta x$) and allowing for crack-free joining at higher processing speeds. Furthermore, the proposed phase-field-ANN framework is significantly more efficient than traditional parameter optimization methods such as Design of Experiments (DOE). While DOE requires costly, time-intensive experimental campaigns to map the process window, the phase-field model provides a deep physical understanding of crack initiation, which is then mapped by the ANN to predict optimal parameters under uncertainty. This hybrid approach reduces experimental optimization time compared to traditional empirical methods.

2. Materials and methodologies

2.1. Materials

The materials selected for this investigation— AA1050 aluminium and BOROFLOAT® 33 (BF33) glass with a thickness of 0.5 mm constitute a benchmark combination for high-reliability hermetic packaging in the electronics and optoelectronics sectors. AA1050 was selected primarily for its high purity (min. 99.5% Al) and superior thermal conductivity, which are essential for heat dissipation in power electronics; its chemical composition and thermo-physical properties are summarized in Tables 1 and 2, respectively. The glass substrate, BF33, was chosen for its exceptional thermal shock resistance and low coefficient of thermal expansion ($3.3\text{x}10^{-6}$ K^{-1}), which minimizes residual stresses during the rapid thermal cycles inherent to laser welding. Due to its chemical durability and optical clarity, BF33 is widely utilized in digital projection, microfluidics, and automotive observation windows. The detailed composition and mechanical properties of BF33 are provided in Tables 3 and 4.

To ensure reproducibility and eliminate interfacial contaminants, both the BOROFLOAT® 33 glass and AA1050 aluminum samples underwent a rigorous pre-welding surface preparation protocol. The aluminum samples were first polished with 1200-grit SiC abrasive paper to remove the natural oxide layer, followed by ultrasonic cleaning in acetone and ethanol for 10 min each to remove oils and surface debris. The glass samples were similarly cleaned ultrasonically in ethanol and then rinsed with deionized water. Finally, both materials were dried in a vacuum oven at 100 °C for 30 min to ensure a completely anhydrous

Table 1
Chemical composition of the AA1050 alloy (wt%) [27].

Al	Fe	Si	Mg	Mn	Ni	Cu	Cr	Zn	Pb
balanced	0.21	0.0693	≤0.05	0.017	≤0.1	0.0823	≤0.03	≤0.015	≤0.03

Table 2
The physical and chemical properties of the aluminum AA1050 [28].

Property	Value	Unit
Density	2.71	g/cm^3
Melting point	933	K
Thermal conductivity	222	W/(m•K)
Thermal Expansion	24 x10^{-6}	K^{-1}
Specific heat	899	J/(kg•K)
Elastic modulus	70	GPa
Shear modulus	26	GPa
Poisson's ratio	0.33	

Table 3
Chemical composition of the glass material (wt%) [29], [30].

SiO_2	B_2O_3	Na_2O/K_2O	Al_2O_3
81	13	4	2

Table 4
The physical and chemical properties of the glass [31].

Property	Value	Unit
Density	2.23	g/cm^3
Transformation temperature	818	K
Softening point	1113	K
Thermal conductivity	1.2	W/(m•K)
Thermal Expansion	3.25 x 10^{-6}	K^{-1}
Specific heat capacity	0.83	kJ/(kg·K)
Elastic modulus	63.1	GPa
Shear modulus	26.7	GPa
Poisson's ratio	0.18	
Fracture Energy	9.35	J/m^2
Internal Length Scale	0.00001	m

surface prior to laser joining.

2.2. Simulation setup

In this study, we employed the COMSOL Multiphysics finite element analysis (FEA) software to develop a three-dimensional model for simulating the temperature distribution around the laser-irradiated area. The entire finite element model was meshed using tetrahedral elements. Fig. 1(a) shows the configuration of the dimensions and the lapped portion. To reduce the simulated time, the lapped portion of the materials was used to simplify the finite element model. Fig. 1(b) shows the non-uniform mesh pattern, which was designed to enhance computational efficiency while maintaining high numerical precision. The laser-irradiated region and the glass–metal interface were modeled with a finer mesh, with a minimum element size of 5 μm. This refinement is necessary for the phase-field method to accurately capture diffuse crack morphologies. In accordance with the standard numerical stability criterion, which requires the mesh size to be no greater than half the internal length scale (l_{int}), the internal length scale was set to 10 μm, ensuring that the dissipation of energy during fracture is mesh-independent and the damage localized within the interface region is physically representative of the cohesive failure modes observed in experimental trials. Furthermore, this ensures that at least two elements span the damage regularization zone, thereby making the results for the damage variable (d) and the local thermal-stress fields independent of the mesh. Additionally, a negative defocus of approximately −0.2 mm was applied to enlarge the irradiation area, promoting more uniform energy distribution and reducing extreme temperature gradients that could otherwise induce early numerical instability.

The damage evolution in BF33 glass is modeled using a phase-field formulation, where the critical energy release rate (G_c) and the internal length scale (l_{int}) are the primary parameters. The parameter values used in this study are presented in Table 4. The fracture toughness was selected based on the SEPB (Single-Edge Pre-cracked Beam) measurements reported by Ref. [30], which provide the most accurate intrinsic values for borosilicate glasses.

To accurately capture the thermomechanical response of the materials under laser irradiation, it is essential to first establish a precise transient thermal field. The evolution of temperature plays a critical role in governing thermal expansion, residual stress development, and damage initiation. Therefore, before coupling the temperature field with the solid mechanics and phase-field fracture modules, we first solve the transient heat conduction problem induced by a moving laser source.

The transient temperature distribution was computed using the heat transfer in solids module, which solves the heat conduction equations incorporating both conduction and convective effects under time-dependent conditions. The governing thermal behavior of the material is described by the following equation as [32]:

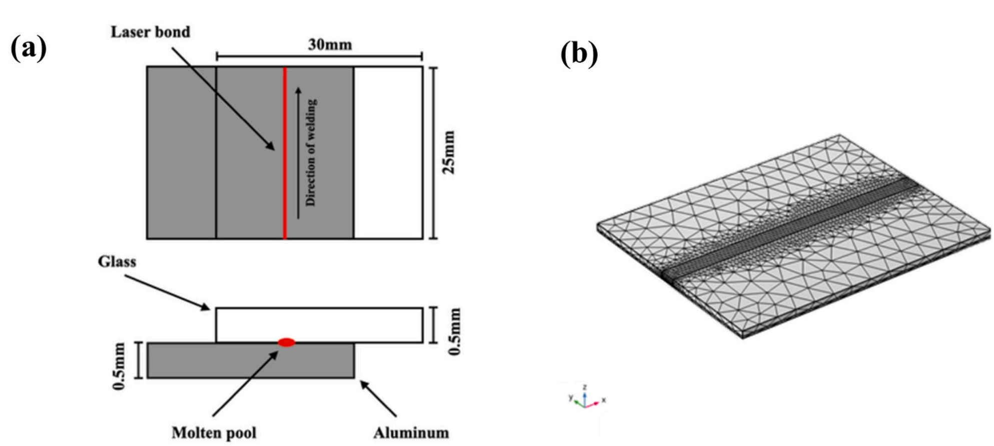


Fig. 1. Schematics of (a) the glass and aluminum dimensions, and (b) the nonuniform mesh pattern used for the geometrical model.

$$\begin{cases} \rho C_p \dfrac{\partial T}{\partial t} + \nabla \cdot \mathbf{q} = Q \\ \mathbf{q} = -k \nabla T \end{cases} \tag{1}$$

where ρ is the density, C_p is the specific heat capacity at constant pressure, T is the temperature field, $\mathbf{q}$ is the heat flux vector, $\nabla \cdot \mathbf{q}$ is the divergence of heat flux, k is the thermal conductivity, and Q is the heat source term. To accurately represent the energy deposition during the welding process, Q is further defined as a combination of surface and volumetric components.

Theoretically, the heat distribution in the glass layer should be represented using a volumetric heat source, while the aluminum layer should be modeled with a surface heat source [33]. Based on Beer--Lambert's law, when a laser beam with a circular spot and a Gaussian intensity profile scans at a constant velocity v, the volumetric heat source q_G in the glass layer can be mathematically expressed as [33]:

$$q_G(x,y,z) = \frac{2(1-R_{AG})\alpha I}{\pi[r_G-(2H-z)tan\ \theta]^2} e^{-\alpha(H-z)} e^{-2\frac{(x-x_0-vt)^2+(y-y_0)^2}{[r_G-(H-z)tan\ \theta]^2}} \tag{2}$$

The thermal input is governed by the incident laser power (I) and the specific optical losses at the air–glass boundary, represented by the reflectivity (R_{AG}). As the beam propagates, the energy distribution is influenced by the absorption coefficient (α), the divergence angle (θ), and the initial spot radius (r_G) on the upper glass surface. Considering the total specimen thickness (H), the effective heat flux reaching the aluminum layer is further modulated by its surface absorptivity (η_s). Consequently, the surface heating source (q_s) for the aluminum layer is defined as follows:

$$q_S(x,y) = \frac{2I\eta_s}{\pi r_A^2} e^{-2\frac{(x-x_0-vt)^2+(y-y_0)^2}{r_A^2}} \tag{3}$$

where r_A is the spot radius on upper surface of the aluminum layer, v is the scanning speed, and x_0 and y_0 are the coordinates of the starting point of the scanning process. We determined the volumetric heat-source parameters based on BF33 glass's absorption of near-infrared light. Our model uses an effective absorption coefficient to capture the semi-transparent nature of borosilicate glass. To check if this approach works, we compared simulated temperature isotherms with experimental values. The close match between the simulated HAZ and the observed results in experiments shows that our model accurately reflects how energy is deposited. This gives us confidence that our calculation of the damage-driving energy density (H_d) is based on a reliable thermal field, which supports accurate phase-field damage predictions.

In natural convection, the boundary heat flux is imposed using a convective heat-transfer boundary condition as [34]:

$$-\boldsymbol{n} \cdot \boldsymbol{q} = q_0 \tag{4}$$

$$q_0 = h(T_{ext} - T) \tag{5}$$

where $\boldsymbol{n}$ is the outward unit normal vector, $\boldsymbol{q}$ is the conductive heat flux vector, q_0 is the prescribed boundary heat flux, T_{ext} is the ambient temperature, and T is the surface temperature. In addition, the convective heat-transfer coefficient h is calculated from the Nusselt number correlation as

$$h = \frac{k}{L} Nu_L \tag{6}$$

where k is the thermal conductivity of the surrounding fluid, L is the characteristic length which been determined as the ratio of the surface area A to the perimeter P of the aluminium plate ($L = A/P$) which is the standard definition for complex, non-circular shapes to accurately represent heat transfer from the upper surface, and Nu_L is evaluated from the Rayleigh number Ra_L using the following piecewise correlations as [34]:

$$Nu_L = \begin{cases} \dfrac{k}{L} 0.54 Ra_L^{\frac{1}{4}}, if\ T > T_{ext}\ and\ 10^4 \le Ra_L \le 10^7 \\ \dfrac{k}{L} 0.15 Ra_L^{\frac{1}{3}}, if\ T > T_{ext}\ and\ 10^7 \le Ra_L \le 10^{11} \\ \dfrac{k}{L} 0.27 Ra_L^{\frac{1}{4}}, if\ T \le T_{ext}\ and\ 10^5 \le Ra_L \le 10^{10} \end{cases} \tag{7}$$

This study incorporates wobbling technology as a key step in the laser welding process. This method effectively minimizes the HAZ while promoting targeted melting. The beam was made to rotate along a circular path with a diameter equal to the wobbling width A and a tangential speed V_t. The center of the circular path was moved along the welding direction at a constant speed V_f, so that two subsequent revolutions achieved a fixed percentage overlap. The laser beam rotates along a circular trajectory with a diameter equal to the swing width A and has a tangential velocity V_t. The center of the circular trajectory moves along the welding direction at a constant speed V_f, so that two consecutive rotations achieve a fixed percentage of overlap. The trajectory of the laser oscillation is shown in Fig. 2, and from Refs. [35–37], the equations can be expressed by:

$$\begin{cases} x(t) = \dfrac{A}{2}\cos\left(\dfrac{2V_t}{A}t\right) + V_f t \\ y(t) = \dfrac{A}{2}\sin\left(\dfrac{2V_t}{A}t\right) \end{cases} \tag{8}$$

The phase of the oscillating trajectory can be expressed as:

$$phase = \frac{2V_t}{A}t \tag{9}$$

The equipment used in the experimental activity was programmed in terms of wobbling amplitude A, overlap O_v, and tangential speed V_t. The wobbling frequency f and welding speed V_f can be calculated with the following equations:

$$f = \frac{V_t}{\pi A} \tag{10}$$

$$V_f = f\left(A - \left(\frac{A \cdot Ov}{100}\right)\right) \tag{11}$$

By differentiating Eq. (8) with respect to time, the velocity of the laser trajectory can be obtained:

$$\begin{cases} \dfrac{dx}{dt} = -V_t \sin\left(\dfrac{2V_t}{A}t\right) + V_f \\ \dfrac{dy}{dt} = V_t \cos\left(\dfrac{2V_t}{A}t\right) \end{cases} \tag{12}$$

From Eq. (12), the laser velocity at different phases is determined, as listed in Table 5.

As shown in Table 5, the laser trajectory velocity exhibits substantial phase dependence. This behavior may introduce spatiotemporal nonuniformity in heat flux and thermal histories, which could adversely affect weld strength and, in more severe cases, increase the likelihood of crack initiation and propagation. To effectively suppress this situation, we used MATLAB to calculate the number of pulses received at each point along the laser path and plotted the results to visualize the pulse distribution, as shown in Fig. 3. We further assessed the distribution using the coefficient of variation (CV) and a uniformity index, enabling a direct evaluation of distribution quality under a given set of process parameters.

Recognizing that the spatial pulse distribution depends on the repetition rate, we maintained a constant scan speed of 5 mm/s and evaluated frequencies ranging from 10 to 100 kHz. For each setting, we computed the CV and a uniformity index and visualized the results to

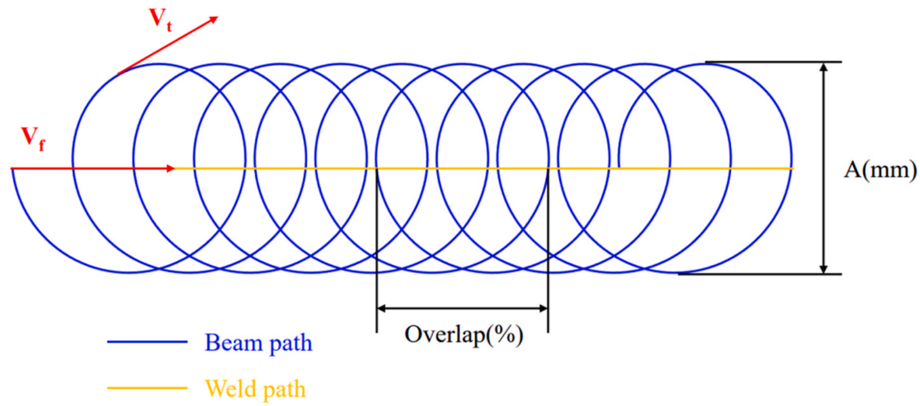


Fig. 2. Schematic of the wobbling strategy.

Table 5
Laser velocity at different phases.

Phase	Velocity
0	$\frac{dx}{dt} = V_f, \frac{dy}{dt} = V_t$
$\frac{\pi}{2}$	$\frac{dx}{dt} = -V_t + V_f, \frac{dy}{dt} = 0$
π	$\frac{dx}{dt} = V_f, \frac{dy}{dt} = -V_t$
$\frac{3\pi}{2}$	$\frac{dx}{dt} = V_t + V_f, \frac{dy}{dt} = 0$

highlight inter-frequency differences. Fig. 4 indicates that, for a fixed scanning speed, decreasing the pulse frequency improves the coefficient of variation (approaching 0) and the uniformity (approaching 1). Therefore, a pulse frequency of 10 kHz is chosen in this study.

To more rigorously assess the likelihood of damage initiation and propagation, we employed a phase-field fracture framework implemented in COMSOL Multiphysics, which enables the prediction of the spatiotemporal evolution of the damage variable and the estimation of failure risk under specified process conditions. Building upon this framework, a fully coupled thermo-mechanical phase-field model was developed to simulate laser-induced cracking behavior in BF33. The phase-field approach provides a concise method for predicting complex crack topologies in glass-metal joining. To maintain computational tractability, this study treats glass as an isotropic material with temperature-independent fracture toughness (G_c).

The fracture process is governed by the minimization of a total en-

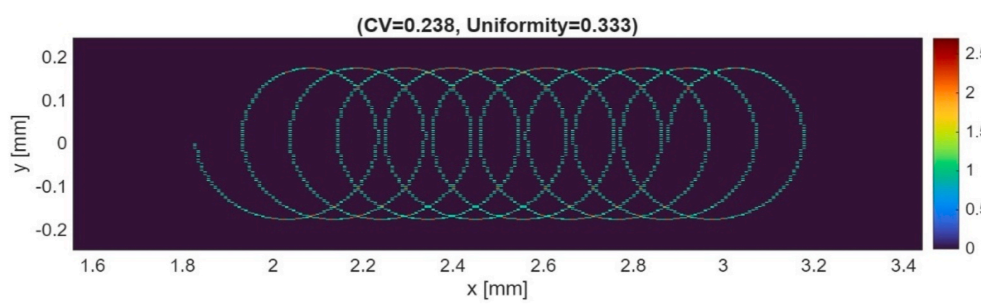


Fig. 3. Pulse distribution visualization.

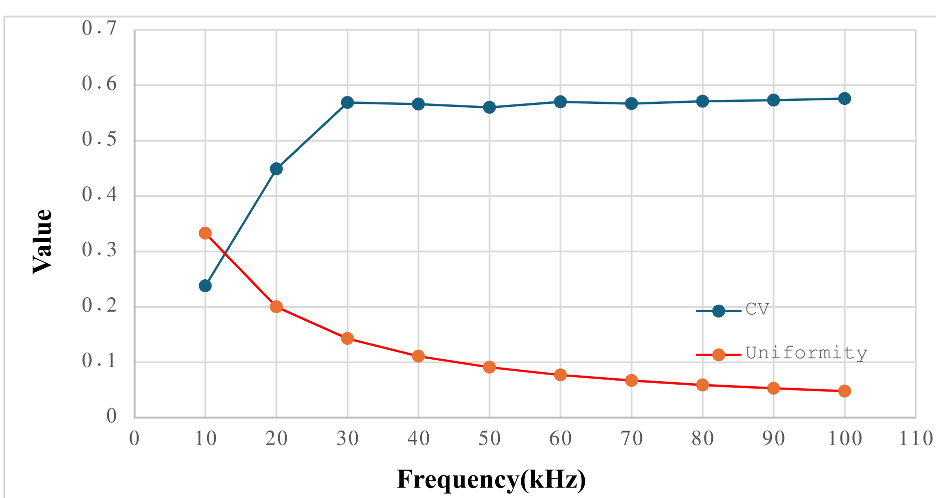


Fig. 4. The CV and uniformity index at different frequencies.

ergy function that incorporates both elastic and fractured energies, which determines the evolution of the system. To elucidate this mechanism, the relevant equations are provided below. In accordance with the general variational phase-field framework described in Ref. [38], the formulation assumes a self-consistent energy balance as follows:

$$\Psi = \int_{\Omega} \left[g(d)\psi_e^+(\epsilon) + \psi_e^-(\epsilon)\right] d\Omega + \int_{\Omega} G_c \left[\frac{(d-1)^2}{4l_0} + l_0(|\Delta d|)^2\right] d\Omega \tag{13}$$

where ψ_e^+ and $\psi_e^-(\epsilon)$ represent the tensile and compressive parts of the elastic energy density, and l_0 is the length scale parameter. The laws of crack motion established in previous research [39] indicate that treating fracture energy as a constant surface energy provides a valid and robust approximation for amorphous materials such as borosilicate glass in the ideal brittle limit.

The assumption of temperature-independent fracture properties is further supported by the thermal-mechanical timeline. In nanosecond-wobbling welding, the primary crack driving force (H_d) comes from significant stress generated by the Coefficient of Thermal Expansion (CTE) mismatch. Fracture initiates during the cooling phase. At this stage, the material has receded from peak temperatures and its properties have stabilized. The model captures the macroscopic crack morphology observed in experiments (Fig. 6). This simplified approach avoids prohibitive computational costs associated with fully coupled temperature-dependent fracture tensors. It retains the predictive capability needed to assess the influence of laser parameters on joint integrity.

The simulation considers the effects of laser heating, thermally induced stress, and fracture propagation driven by damage accumulation. The temperature field is first calculated using the time-dependent heat transfer in solids module, which serves as the thermal load input for the solid mechanics module. Thermal expansion is described by the constitutive relation:

$$\varepsilon_{th} = \alpha(T)\left(T - T_{ref}\right) \tag{14}$$

where α(T) is the temperature-dependent thermal expansion coefficient and T_{ref} is the stress-free reference temperature. This thermal strain generates internal stress during rapid localized heating, which contributes to the initiation of damage in the brittle material. The damage evolution is governed by the phase-field formulation based on the energy minimization approach, expressed as:

$$\frac{\partial Q}{\partial d} - l_{int}^2 \nabla\cdot(D\nabla d) = \frac{\partial \phi}{\partial d} H_d \tag{15}$$

where d is the scalar phase-field variable ranging from 0 (intact) to 1 (fully damaged), l_{int} is the internal length scale controlling the damage zone width, and H_d is the damage history field defined as:

$$H_d = \max\left(D_d, H_{d,old}\right) \tag{16}$$

where $D_d$denotes the damage-driving energy density computed at the current time step, and $H_{d,old}$ represents the previously stored history field that records the maximum value attained up to the preceding time step. The degradation function is defined as [40]:

$$\phi(d) = (1-d)^2 \tag{17}$$

which ensures a smooth degradation of material stiffness as damage increases. Mechanical stress is modified accordingly by:

$$S_{dmg} = (1-d)S \tag{18}$$

where S is the stress from the undamaged solid. The governing equations for heat transfer, solid mechanics, and phase-field fracture are solved in a fully coupled manner, with damage affecting the stress field and vice versa. This approach allows the model to capture the interplay between rapid thermal gradients, mechanical response, and crack propagation, which is critical for understanding and optimizing the laser processing of brittle materials.

2.3. Experimental setup

Lap joints were fabricated using a nanosecond ytterbium fiber laser (IPG Photonics YLPN-1000-4 × 200-30-M, λ = 1064 nm) as shown in Fig. 5. The pulse repetition frequency (10 kHz) and pulse duration (50 ns) were optimized to provide sufficient energy density for robust interfacial bonding. The optical train was configured to achieve precise surface focusing, employing a 125 mm focal length, a 15 μm spot diameter, and beam quality $M^2 < 1.5$. Parameter settings were established through preliminary trials to ensure process stability and weld reproducibility, thereby supporting the reliability of the reported results. To interrogate process sensitivity, the scanning speed and average laser power were systematically varied and their effects on weld formation and joint performance were evaluated.

During the welding process, the laser beam was focused perpendicular to the top surface of the aluminum foil and rotated along a circular path with a wobble diameter of d_w at a tangential velocity of V_t. The center of this circular path moved along the welding direction at a constant speed, ensuring that successive beam revolutions overlapped by 70%. The pulsed laser mode employed a trapezoidal power profile, with key parameters being the peak laser power (W), pulse frequency (kHz), and pulse duration (ns).

In Table 6, various combinations of peak laser power and scanning speed were tested using the wobbling strategy. For all trials, the pulse duration was set to 50 ns, and the pulse frequency was 10 kHz. The selection of these parameters followed the principles of design of experiments (DOE), in which the scanning speed (v) and peak power were defined as the main factors. The feasible range was first determined through preliminary trials to avoid under-melting or excessive spatter. We then selected discrete levels of scanning speed (2–6 mm/s) and peak power (1.875–5.625 kW), corresponding to single-pulse energies of approximately 93.75–281.25 μJ at a fixed pulse duration of 50 ns and a fixed repetition rate of 10 kHz. To isolate the effects of v and P_{peak}, the frequency was kept constant; the resulting average power therefore spanned 0.94–2.81 W ($\bar{P} = E_p \times f$). The experiments were executed in randomized order to minimize systematic bias, and each condition was repeated to ensure reproducibility and statistical validity.

3. Result of simulation and experiment

3.1. Simulation results

Fig. 6 illustrates the phase-field simulation results for damage evolution and the corresponding stress distribution for two laser power conditions (60W and 40W) at a constant scanning speed of 2 mm/s. Thermal analysis of the 60W condition reveals a peak interface temperature reaching approximately 1248 °C. This temperature significantly exceeds the melting point of the AA1050 alloy, facilitating a robust melt pool, but also creates a steep thermal gradient relative to the glass substrate. At higher laser power (Fig. 6(a)), the thermal input is more intense, resulting in a wider damage-affected zone, as captured by the phase-field variable. The maximum damage variable reaches approximately 0.5, indicating severe material degradation near the fusion boundary. The corresponding stress field (Fig. 6(b)) shows a high-stress concentration in the vicinity of the fusion zone, particularly at the HAZ boundary, where the stress exceeds 10 MPa which is a direct consequence of the steep thermal gradient; the rapid contraction of the molten aluminum during solidification against the rigid, non-expanding glass superstrate generates significant tensile forces potentially serving as a crack initiation site.

In contrast, at lower laser power (Fig. 6(c)), the peak temperature is

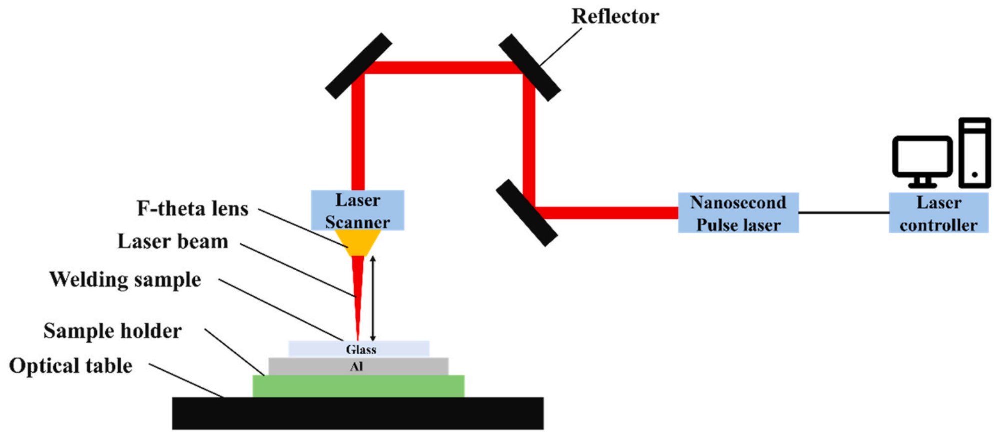


Fig. 5. Schematics of the laser welding setup.

moderated to 900^{0}C both the extent of damage and the magnitude of stress are significantly reduced. The phase-field results indicate that the damage variable is more localized, with a peak at approximately 0.16, while the stress field (Fig. 6(d)) exhibits a shallower gradient, reaching a peak stress of around 5 MPa. This suggests that reduced power conditions can help suppress damage initiation but may also compromise joint penetration and bonding quality.

Overall, these results confirm that the magnitude and spatial distribution of thermal stress strongly influence damage accumulation during laser welding. The phase-field model effectively captures the dynamic evolution of damage driven by thermomechanical interactions, allowing for quantitative assessment of process parameters on joint integrity.

3.2. Experimental results

The figure presents optical microscopy (OM) images of the top view after laser processing at average powers of 40 W and 60 W, with a pulse frequency of 10 kHz, highlighting differences in melting behavior and the HAZ. At 40 W (Fig. 7(a)), the molten trace exhibits a relatively narrow and uniform morphology, indicating localized melting with limited thermal diffusion. The periodic ripple patterns along the melt track suggest consistent pulse overlap and stable surface flow under moderate energy input.

In contrast, at 60 W (Fig. 7(b)), the molten region becomes considerably wider and darker, accompanied by a rougher surface texture. The darker appearance is primarily due to the formation of pores within the glass layer, which scatter light and reduce surface reflectivity. This observation implies intensified melting, increased vaporization, and a broader HAZ due to higher thermal accumulation and energy density.

These results indicate that variations in laser power and scanning speed have a significant impact on the melting morphology, HAZ, and defect formation. The observed pore-induced darkening and roughened surfaces at higher power confirm the accumulation of thermal stress within the glass–metal interface. Such thermally induced stress and localized vapor pressure are believed to play a decisive role in crack initiation during solidification. Therefore, to further elucidate the relationship between these defects and interfacial fracture behavior, a phase-field-based crack analysis is conducted in the following section.

The optical characteristics of the joint interface were evaluated using UV-Vis transmittance spectroscopy, which confirmed the macroscopic darkening observed. To distinguish the effects of laser welding from inherent glass properties, an intra-sample control was implemented using a Hitachi U4100 spectrophotometer. Spectra were obtained from both the untreated 'Bare Glass' and the localized 'Weld Point' of each specimen. Fig. 8 indicates that pristine BF33 glass exhibits high transparency, with transmittance values of 95–96% at 550 nm.

In the optimized 'Green Zone' (40 W, 2 mm/s, Fig. 8(a), the transmittance at the weld point is nearly identical to that of the bare glass, indicating the preservation of optical integrity. Conversely, in the high-energy 'Red Zone' (60 W, 2 mm/s, Fig. 8(b)), the weld point exhibits a significant reduction in transmittance, decreasing to 75%, whereas the transmission spectra. This ~20% optical attenuation is attributed to Mie scattering and diffuse reflection caused by a high spatial density of interfacial micro-pores and voids. To ensure accurate measurement despite the opaque aluminum substrate, the incident beam was precisely aligned with the glass superstrate, thereby capturing the light extinction caused by the interfacial morphology before it reached the metal backing. These quantitative results demonstrate that the observed darkening is a localized physical phenomenon arising from the weld's porous structure rather than chemical degradation of the glass.

4. Crack analysis

4.1. Cracking criterion for laser welding of BF33–Al

In this study, the cracking criterion for BF33–Al joints is defined purely within a phase-field damage framework that quantitatively captures crack initiation and propagation under laser-induced thermo-mechanical stresses. Fracture is represented by a continuous damage variable $d(\mathrm{x},t) \in [0,1]$, where $d=0$ denotes intact material, and $d=1$ represents a fully fractured zone; material stiffness degrades smoothly via the standard function $\phi(d) = (1-d)^2$. Damage evolution is governed by the damage-driving energy density D_d and its history field $H_d = \max(D_d, H_{d,\mathrm{old}})$; once the critical energy release rate G_c is exceeded locally, damage grows, and cracks advance [41,42].

To account for laser-induced thermal loading, the phase-field formulation is thermo-mechanically coupled to the transient temperature field. Thermal strain follows Eq. (11), $\varepsilon_{\mathrm{th}} = \alpha(T)\left(T - T_{\mathrm{ref}}\right)$, and the CTE mismatch between BF33 glass and aluminum drives interfacial residual tensile stresses that accumulate in H_d and promote damage growth. Within this energy-based criterion, fracture initiates when the local condition $H_d \geq G_c$ is met, equivalently, when d approaches a calibrated threshold $d_{\mathrm{crit}} \approx 0.7$ in our case [43,44]. It is noted that crack initiation and propagation in BF33 glass are determined by the history field of the strain energy density, H_d, which serves as the driving force for the damage variable d. According to the thermodynamic framework for brittle fracture [45], the total elastic strain energy density (ψ_ε) is separated into tensile (ψ_ε^+) and compressive (ψ_ε^-) components to emphasize that brittle glass failure is primarily driven by tensile loads [45]:

$$\psi_{\varepsilon}{}^{+}(\varepsilon) = \lambda/2\langle tr(\varepsilon)\rangle_+^2 + \mu tr\left(\varepsilon_+^2\right) \tag{19}$$

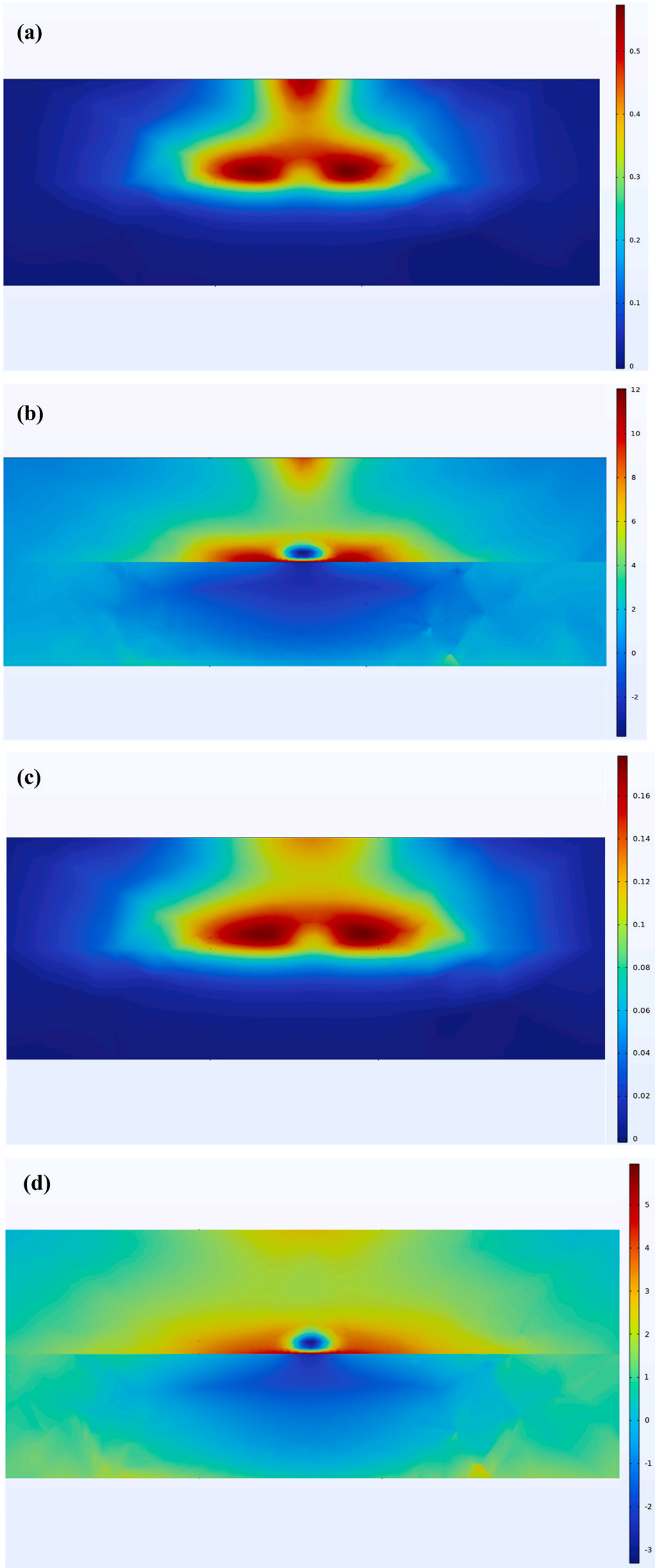


Fig. 6. Simulation results of laser welding under different power conditions at a constant scanning speed of 2 mm/s (a) and (b) show phase-field damage variable and stress distribution at 60W laser power; and (c) and (d) show corresponding results at 40W laser power.

Table 6
The laser parameters selected in welding.

Number	Scanning speed (mm/s)	Frequency (kHz)	Peak power (W)	Pulse Energy (μJ)
1	2	10	1875	93.75
2	2	10	2500	125
3	2	10	3125	156.25
4	3	10	2500	125
5	3	10	3125	156.25
6	3	10	3750	187.5
7	4	10	3125	156.25
8	4	10	3750	187.5
9	4	10	4375	218.75
10	5	10	3750	187.5
11	5	10	4375	218.75
12	5	10	5000	250
13	6	10	4375	218.75
14	6	10	5000	250
15	6	10	5625	281.25

$$\psi_{\epsilon}^{-}(\varepsilon) = \lambda/2\langle tr(\varepsilon)\rangle_{-}^{2} + \mu tr(\varepsilon_{-}^{2}) \tag{20}$$

where ε_+ and ε_- are tensile and compressive strain tensors. In laser welding, the strain tensor ε represents both the immediate thermal expansion mismatch and the resulting residual stress field. Laser heating at the aluminum-glass interface causes a substantial CTE mismatch, which generates significant localized thermal stresses. As the material cools, these stresses become residual stresses, which are subsequently modeled by coupling thermal elasticity with the phase-field history variable. To avoid artificial crack healing during the cooling cycle, H_d is defined as the maximum tensile strain energy density reached during the entire processing period as [45]:

$$H_d(x,t) = \max_{s\in[0,t]} \psi_{\epsilon}^{+}(\varepsilon(x,s)) \tag{21}$$

where x represents the spatial position in the finite element analysis (FEA) mesh, t is the current simulation time step, and s tracks the loading cycle from the start ($s = 0$) to the current time ($s = t$). Applying the maximum operator over [0, t] ensures that the damage-driving force H_d increases monotonically. This method captures the highest thermal and residual energy states and prevents unrealistic crack 'healing' during cooling [46]. The damage-driving energy density, H_d, is fundamentally coupled to the transient thermal cycle. For the representative 40 W case, the FEA model predicts a peak interface temperature of 900 °C. This thermal excursion generates a localized expansion mismatch between the BF33 glass and the aluminum substrate. As the joint cools, the resulting tensile strain components are processed through the elastic energy functional ψ_{ϵ}^{-}, where the Lamé constants (λ, μ) translate the thermal-elastic strain into the history-maximum driving force. Despite the high peak temperature, the magnitude of H_d at 40 W remains below the critical energy release rate G_c. This ensures that the cracking criterion $H_d \geq G_c$ is not satisfied, theoretically validating the high-quality, fracture-free 'Green Zone' observed in the experiment shown in Figs. 12 and 16.

Parametric simulations indicate that higher laser power with slower scanning increases heat accumulation, elevates H_d, and broadens the damage-affected zone; conversely, moderate power with faster scanning lowers the accumulated driving energy and confines damage spatially. Consequently, the proposed phase-field cracking criterion enables a physically consistent prediction of failure without relying on geometric or empirical surrogates, and it provides a direct link between process parameters (e.g., laser power, scanning speed) and damage evolution. The simulated damage fields (d-maps) agree closely with our experimental results, confirming the model's reliability in capturing crack initiation and growth in BF33–Al laser joints.

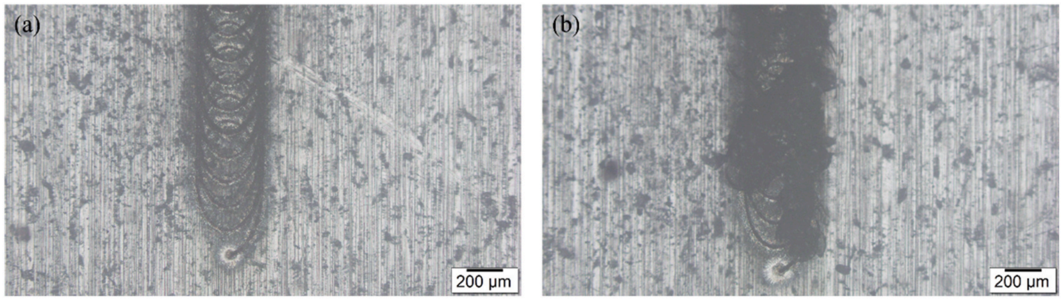


Fig. 7. Comparison between two different samples for the top side of BF33 and Al for (a) Avg. laser power of 40 W and scanning speed of 2 mm/s, and (b) Avg. laser power of 60 W and scanning speed of 2 mm/s, respectively.

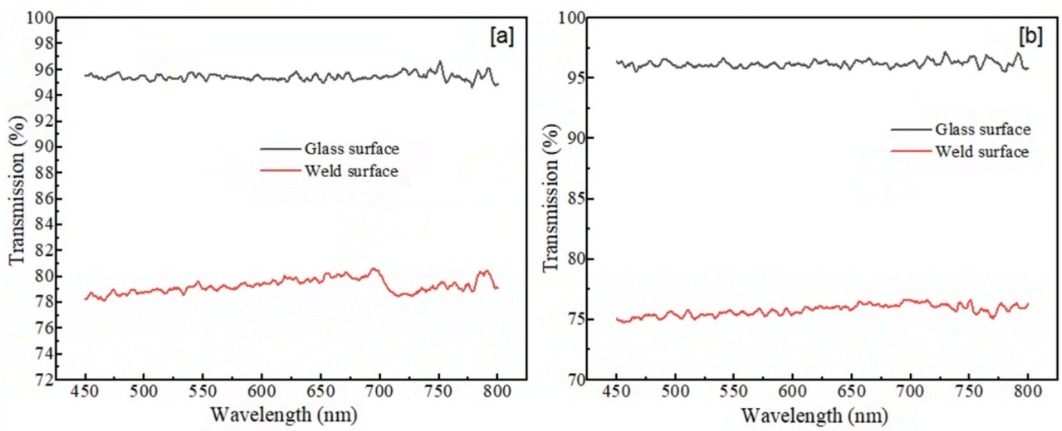


Fig. 8. Comparative UV-Vis transmittance spectra of BF33 glass superstrates under different processing conditions: (a) Comparison of glass with the optimized weld interface (40 W, 2 mm/s), indicating the preservation of high optical throughput; and (b) Comparison with the high-energy regime (60 W, 2 mm/s), demonstrating a significant attenuation in transmittance across the visible spectrum.

4.2. *Optimized processing map of BF33–Al*

Section 4.2 presents a multi-stage methodology for developing the optimized welding map for the BF33–Al system. Section 4.2.1 details the application of a circle-packing model to systematically explore the parameter space defined by scanning speed and laser power. The optimized parameters are subsequently used as inputs for phase-field simulations, enabling the construction of a predictive artificial neural network (ANN) framework to model complex damage variables. Section 4.2.2 then delineates the final optimized process boundaries, emphasizing parameter adjustments to minimize thermally induced cracking. The resulting map functions as a localized design tool, specifically adapted to the optical and material constraints of this study.

4.2.1. *Circular packaging design model for parameter optimization*

To construct the process map for BF33–Al laser welding, we employed an ANN-based modeling approach similar to that used in previous studies [47,48]. As shown in Fig. 9, the circle-packed design space spans two key parameters: the average laser power (x-axis) and the scanning speed of the nanosecond laser with a wobbling strategy (y-axis), both normalized to [0, 1]. These correspond to scanning speeds of 1–7 mm/s and average laser powers of 30–90 W. In this work, a MATLAB-based circle-packing algorithm generated 36 sample points, representing diverse combinations of tangential speed and average power, ensuring uniform coverage with a circle radius of 0.075. These samples served as inputs to the phase-field simulations to compute damage levels.

The simulation data were then used to train three ANN models, each predicting the phase-field variables for a given combination of scanning speed and laser power. The trained models subsequently evaluated 2296 parameter combinations over the design space, using increments of 0.15 mm/s (speed) and 1 W (power). A systematic analysis of the 36 critical parameters influencing weld outcomes identified a robust parameter window, thereby enhancing prediction reliability and supporting the construction of the process map.

These process maps visualize the relationships between laser-welding parameters and key output characteristics, providing a structured framework for optimizing BF33–Al bonding and minimizing defects.

4.2.2. *Final optimized processing map*

The ANN models used in this study were developed and trained with MATLAB's Neural Network Toolbox to construct the final optimized processing map for BF33–Al laser welding. These models are designed to predict key phase-field variables based on welding parameters such as average laser power and scanning speed. After training, the ANN models were applied to systematically analyze the processing map and identify an optimal parameter window, thereby ensuring the integrity of BF33–Al joints and minimizing crack-related defects.

The regression plots in Fig. 9 show strong agreement between ANN outputs and targets across training, validation, and test sets (all $R \approx 0.97$–0.99), indicating high fidelity and good generalization. The ANN-based processing map evaluates a dense grid of power–speed pairs and reveals a clear trend: the predicted damage increases with average laser power and decreases with scanning speed, resulting in a low-damage band toward the lower-power and higher-speed side of the map.

The experimentally tested samples, marked in Fig. 10, were examined using optical microscopy to validate the ANN–phase field predictions. Three representative weld morphologies were consistently observed and can be interpreted as distinct thermo-fluid regimes

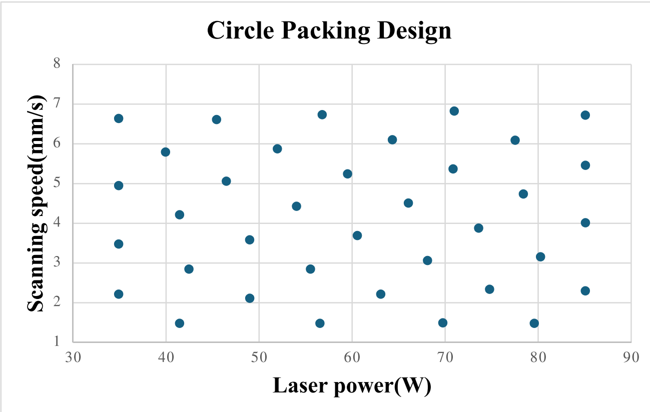


Fig. 9. Circle packing design for Simulation.

governed by the linear energy input (increasing with higher average power and/or lower scanning speed) as.

(i) Green-zone morphology (stable conduction-mode welding): a continuous and uniform weld seam is formed with clearly visible periodic ripples, while the glass surface remains intact and free of surface-breaking cracks. This indicates that the thermal gradient and residual tensile stress in BF33 are sufficiently controlled so that damage remains limited.

(ii) Blue-zone morphology (porosity-dominated without macroscopic cracking): the weld seam remains continuous, but localized darkening and roughened features appear due to subsurface micro voids/pores generated inside the glass. These pores are attributed to enhanced evaporation and transient keyhole-like instability under moderately increased heat accumulation. Although internal defects are present, no macroscopic cracks are observed on the glass surface, suggesting that the driving force is not yet high enough to trigger unstable crack propagation.

(iii) Red-zone morphology (excessive energy input with cracking): the processed track becomes significantly wider and darker with blurred ripples, accompanied by extensive pore formation and visible cracks. This regime is associated with severe thermal accumulation, strong vapor recoil pressure, and amplified tensile residual stresses during cooling, which collectively promote pore coalescence and crack initiation/propagation in BF33.

Based on these characteristic morphologies, the processing map in Fig. 11 is divided into three corresponding regions, demonstrating that the experimentally observed defect modes agree well with the ANN-predicted damage regimes. It is noted that each color represents a distinct regime of the predicted phase-field damage variable (d): (i) the green zone corresponds to $d < 0.3$, indicating a crack-free glass–metal interface; (ii) the blue zone represents $0.3 \leq d < 0.85$, where microvoids or pores form within the glass but no macroscopic cracking occurs; and (iii) the red zone indicates $d \geq 0.85$, signifying severe damage with both porosity and interfacial cracking.

5. Verification in experiments and analysis

5.1. Verification of optimized processing map

To verify the predictive processing map, representative parameter

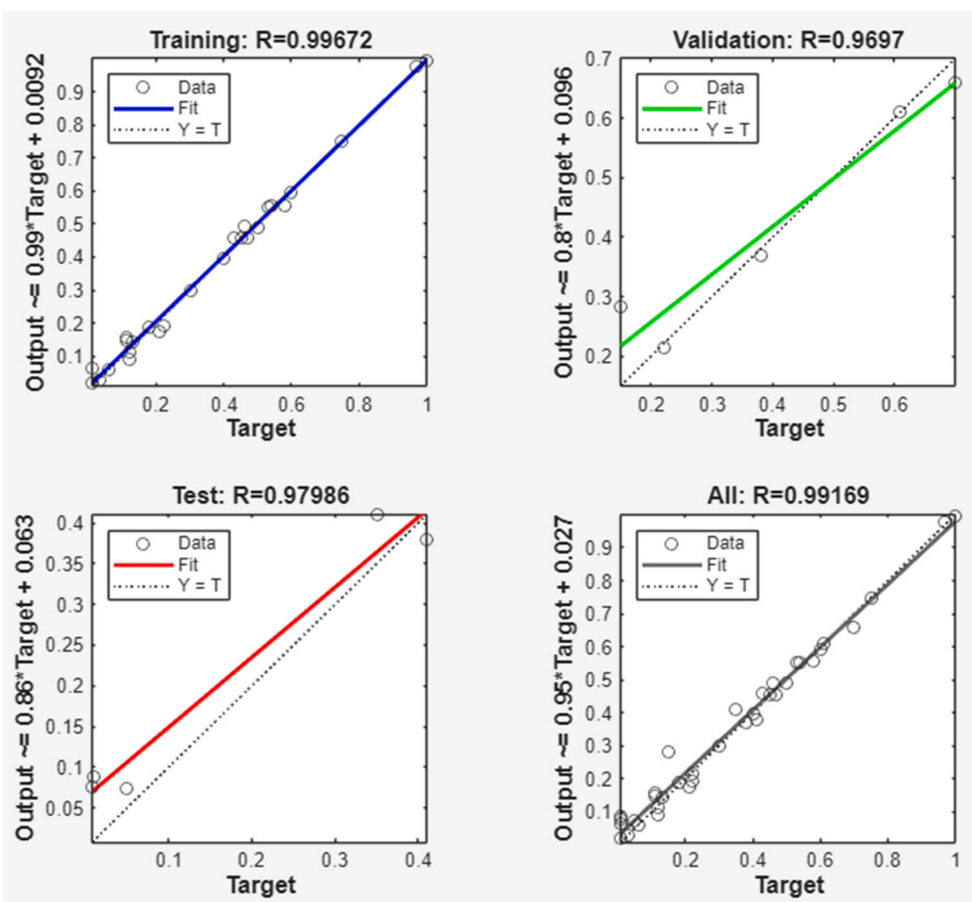


Fig. 10. ANN regression results for the phase-field target.

sets were selected from three characteristic regions and implemented in welding trials. In Fig. 11, the green zone corresponds to a low-damage window and was sampled by A and B. The blue zone represents intermediate energy input, where internal defects may initiate without catastrophic fracture, and was sampled by C and D. The red zone indicates a high-damage regime prone to severe melting and cracking, and was sampled by E and F. The specific laser powers and scan speeds were listed in Table 7.

Under the green zone conditions (Samples A and B), the melt track remained narrow, the HAZ was limited, and no surface-breaking cracks were observed. This morphology is characteristic of stable conduction-mode melting with efficient heat dissipation at moderate power and relatively high scanning speed. In the blue zone (Samples C and D), the melt width and penetration increased, and the overlapping ripples became more pronounced, reflecting the higher linear energy input. Local darkened areas appeared near the fusion boundary, indicating the onset of subsurface porosity or localized keyhole instability, although no macroscopic cracks were detected. Under red-zone conditions (Samples E and F), the excessive energy density produced a wide, deeply penetrating seam with a uniformly dark appearance and blurred ripples, features typical of sustained keyhole behavior and strong evaporation-driven recoil pressure. These characteristics are consistent with an increased tendency for pore coalescence and crack formation, in agreement with the model predictions.

Overall, the experiments corroborate the model-derived processing map. Damage severity increases with higher power and lower scanning speed, whereas defect-free seams are more likely at lower power or higher scanning speed within the green zone.

5.2. Shear strength analysis

To validate the ANN-assisted processing map and quantitatively assess the bonding performance, shear tests were conducted on BF33–Al assemblies using a Nordson DAGE 4000Plus die shear tester (200 kg capacity). A constant blade feed rate of 20 μm/s was employed to minimize dynamic loading on the brittle glass and to ensure that the measured values reflected the intrinsic interfacial strength. In this study, shear tests were performed on samples fabricated within the green zone of the processing map, since this regime corresponds to low predicted damage and crack-free seams. The tested laser parameters and the resulting shear strengths are listed in Table 8. The maximum shear strength reached 8.5 MPa for the sample processed at 40 W and 3 mm/s, while all green zone samples-maintained strengths above 5.6 MPa, confirming the robustness and reproducibility of the optimized window. A measured shear strength of 8.5 MPa is considered competitive for glass–aluminum micro-packaging joints. Previous studies on glass–aluminum bonding and sealing have reported strong hermetic joints and, in some cases, substantially higher strengths, suggesting that 8.5 MPa falls within a credible engineering range [49]. The observed bond aligns with oxide-mediated interfacial reactions, including Al_2O_3- and Al–Si–O-based phases commonly reported in glass–aluminum joining. Furthermore, oscillating or tailored beam strategies have been shown to improve heat-input distribution and reduce defect formation, which correlates with enhanced post-weld mechanical reliability in the absence of chemical surface functionalization.

Table 9 presents a comparative summary of published results on glass–metal laser welding. Direct comparison of absolute shear strength is limited by differences in sample geometry, material grades (for example, Al6061 versus AA1050), and laser modalities. For example, millisecond-laser welding of borosilicate glass to titanium and Al6061 yields shear strengths of 3.09–3.54 MPa and typically requires pre-processing, such as micro-arc oxidation (MAO) [11], to enhance energy absorption. In contrast, the nanosecond-wobbling process reported in this study achieves 8.5 MPa at a lower power (40 W) in an interlayer-free configuration, thereby eliminating the need for chemical surface modifications.

Compared to ultra-short-pulse regimes, the nanosecond-wobbling method maintains high throughput. Femtosecond (fs) welding of Al6082 to quartz achieves 8.94 MPa [26], but requires a very low scanning speed (0.1 mm/s) to generate sufficient localized heating. The nanosecond-wobbling framework attains comparable strength at a scanning speed thirty times higher (3 mm/s). The primary advantage of the nanosecond-wobbling approach is not only its comparable shear strength but also its capacity to address thermal-mechanical property

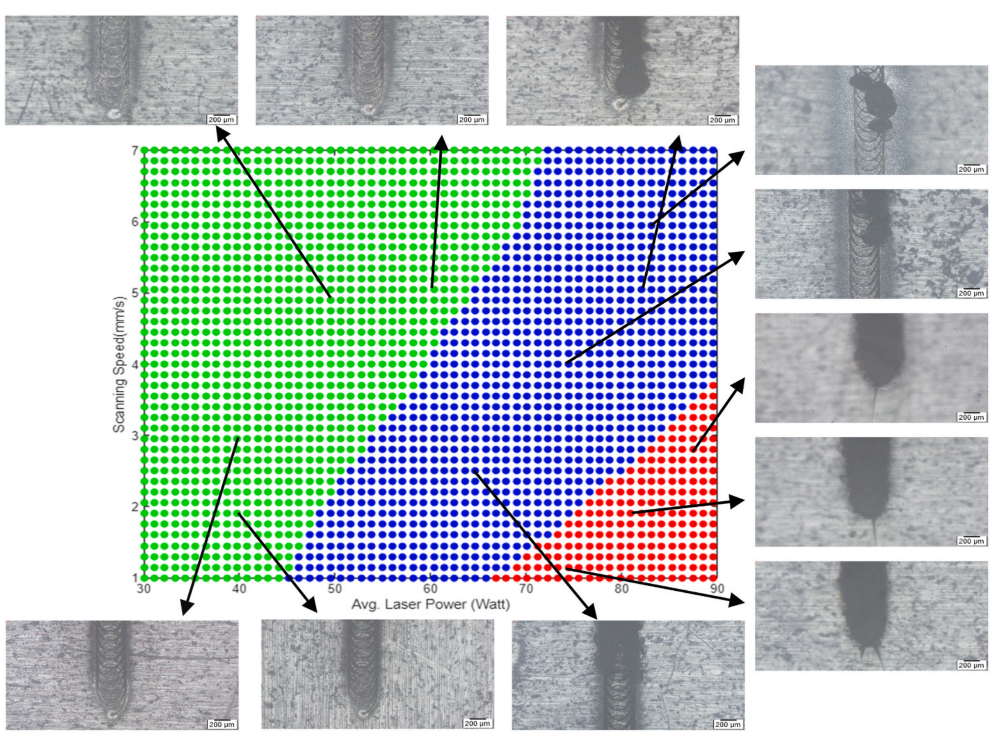


Fig. 11. ANN-derived processing map of predicted damage vs. average power and scanning speed.

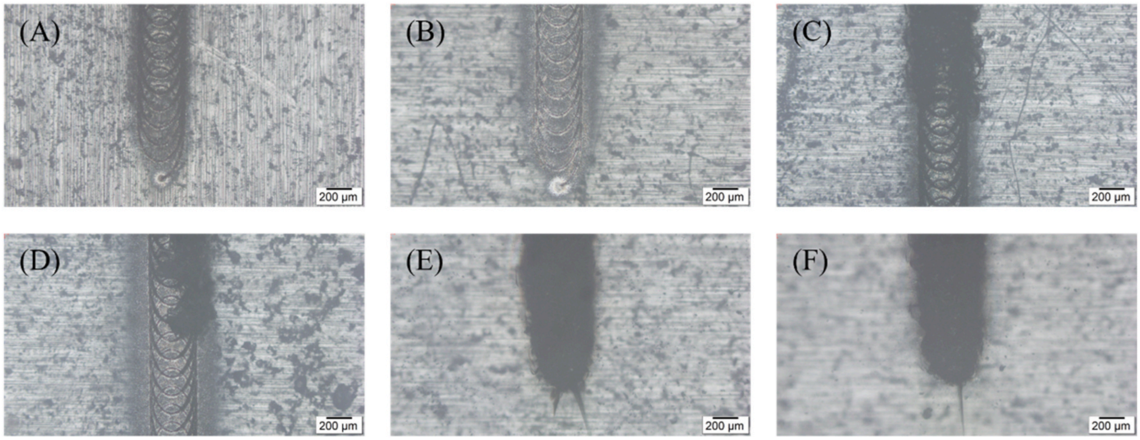


Fig. 12. Optical micrographs of tracks produced under the verification parameters in Table 7.

Table 7
Experimental parameters chosen for validation trials.

Zone	Sample	Power (W)	Scanning speed (mm/s)
Green	A	40	3
	B	53	5
Blue	C	64	2.5
	D	73	4
Red	E	75	1.3
	F	83	2

Table 8
Laser parameters for shear strength testing of BF33–Al bonded samples.

Sample	Power(W)	Scanning speed (mm/s)	Shear strength (MPa)
1	30	2	7.7
2	30	3	5.6
3	40	3	8.5

Table 9
Comparative study of laser welding for glass-metal joints.

Materials	Laser	Power (W)	Scanning speed (mm/s)	Shear strength (MPa)	Ref.
Al/borosilicate glass	**ns**	**40**	**3**	**8.5**	**This work**
Titanium alloy/borosilicate glass	ms	150	3	~3.09	[50]
Al6061/ultra-thin glass	ms	100	3	~3.54	[11]
Kovar alloy/Soda-lime glass	fs	4.5	1	2	[51]
Al6082/quartz glass	fs	5	0.1	8.94	[26]

mismatches through systematic optimization, without the need for complex interlayers or low-productivity laser regimes.

Variations in shear strength, reaching up to 8.5 MPa, are primarily attributed to differences in chemical bonding at the interface. During nanosecond laser processing, elevated local temperatures disrupt Si–O bonds in borosilicate glass, enabling oxygen migration toward the exposed aluminum surface. Consequently, a thin Al–*O*–Si transition layer forms, serving as the principal chemical linkage between the two materials.

The relationship between this chemical process and the resulting mechanical properties is complex. At an optimal energy density of 40 W and 3 mm/s, a smooth and continuous Al–*O*–Si layer develops, resulting in maximum shear resistance. Increased heat input leads to a thicker transition layer, which may transform into brittle complex oxides or entrap small air pockets, as indicated by the reduced transmittance discussed in Section 3.2. This transformation from a robust chemical bridge to a brittle layer account for the observed decrease in shear strength at elevated temperatures. In addition, mechanical integrity of the nanosecond-wobbling weld was evaluated through shear testing on a representative specimen optimized at 40 W and 3 mm/s. The measured shear strength of 8.5 MPa serves as a benchmark for assessing the bonding efficiency of the proposed interlayer-free method.

Post-fracture analysis revealed a predominantly interfacial failure mode, with microscopic evidence of glass remnants localized on the aluminum surface. These observations indicate a transition toward cohesive failure at the nanometer scale, governed by the Al–*O*–Si chemical transition layer. While this preliminary evaluation demonstrates the high strength potential of the process, ongoing research aims to provide a comprehensive statistical characterization across a larger sample population to further evaluate process reliability.

Overall, these results highlight several advantages of the proposed wobbling nanosecond-laser strategy. First, it enables direct, interlayer-free bonding of BF33–Al with shear strengths that not only surpass those of existing millisecond-laser glass–metal joints but also approach those of the best-performing femtosecond-laser systems. Second, the combination of moderate power and relatively high scanning speed provides a favorable balance between mechanical strength and process efficiency, which is crucial for industrial scaling. Finally, the close agreement between the green-zone predictions of the ANN–phase-field processing map and the experimentally measured shear strengths confirms that the proposed numerical framework can reliably guide parameter selection to achieve high-strength, low-damage glass–metal joints.

5.3. SEM & EDS analysis

The effect of the three process regimes on weld morphology is illustrated in Fig. 13, which shows SEM top views of samples processed in the green, blue, and red zones. Under the green-zone condition (Fig. 12(a)), the melt track is relatively narrow and well confined, with clearly defined overlapping ripples and only limited spatters around the processed area. The surrounding surface remains smooth, indicating stable conduction-mode melting and effective heat dissipation at moderate power and higher scanning speed.

For the blue-zone condition (Fig. 12(b)), the molten region becomes markedly wider and the overlapping ripples are thicker and more irregular. A larger amount of resolidified material accumulates along the track edges and on the surface, and the adjacent area appears

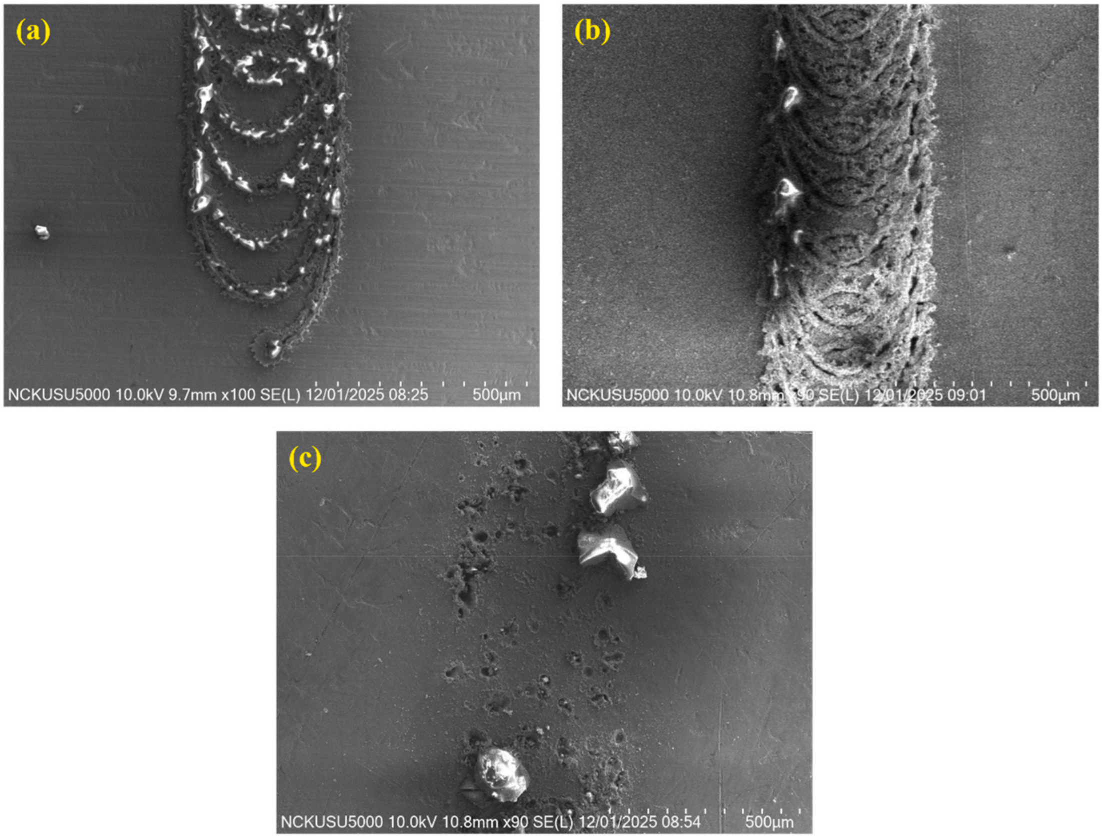
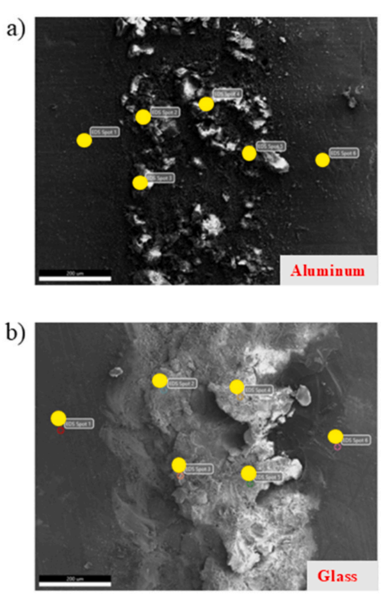


Fig. 13. SEM top-view images: (a) Green zone Sample A; (b) Blue zone Sample C; (b) Red zone Sample F.

roughened, which suggests a higher linear energy input and the onset of a less stable, partially keyhole-like regime.

In contrast, the red-zone condition (Fig. 12(c)) produces only discontinuous agglomerates of resolidified droplets accompanied by numerous pits and surface damage along the scan path. The continuous weld seam observed in the green and blue zones is no longer present, implying excessive energy density and strong evaporation-driven material removal rather than controlled melting.

Based on the characteristic morphologies observed by SEM, the Green Zone (Fig. 14) and Blue Zone (Fig. 15) conditions were selected for localized chemical characterization via EDS to evaluate the bonding mechanism. In the Green Zone, the EDS results (Fig. 14) confirm a high degree of elemental diffusion and chemical hybridization at the interface. On the Aluminum side (Fig. 14(a)), spots within the weld track (e.g., Spots 2, 3, and 5) reveal a significant migration of glass-derived elements. Notably, Spot 3 shows oxygen (O) content of 66.2 at% and silicon (Si) content of 26.3 at%, despite being located on the aluminum substrate. Conversely, on the Glass side (Fig. 14(b)), aluminum is clearly



a)

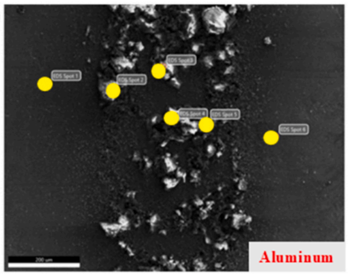
Points	O (at %)	Na (at %)	Al (at %)	Si (at %)	K (at %)
EDS Spot 1	3.7	0.0	96.0	0.1	0.1
EDS Spot 2	30.7	1.4	18.5	45.0	4.4
EDS Spot 3	66.2	2.8	3.9	26.3	0.7
EDS Spot 4	17.5	1.7	8.8	65.2	6.9
EDS Spot 5	33.9	2.2	12	47.8	4
EDS Spot 6	1.2	0.5	97.9	0.1	0.3

b)

Points	O (at %)	Na (at %)	Al (at %)	Si (at %)	K (at %)
EDS Spot 1	60.0	2.1	1.5	35.8	0.5
EDS Spot 2	22.4	1.1	54.8	21.0	0.6
EDS Spot 3	5.7	0.8	87.7	5.1	0.8
EDS Spot 4	53.4	3.0	5.4	37.4	0.7
EDS Spot 5	21.5	2.4	16.8	58.0	1.3
EDS Spot 6	59.6	1.9	1.4	36.6	0.5

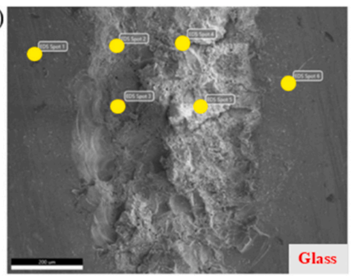
Fig. 14. EDS point analysis of the bonding interface for Sample A (Green zone).

a)



Points	O (at %)	Na (at %)	Al (at %)	Si (at %)	K (at %)
EDS Spot 1	2.5	0.1	97.1	0.1	0.2
EDS Spot 2	13.8	0.2	9.6	70.4	6.0
EDS Spot 3	60.8	2.2	10.7	25.8	0.5
EDS Spot 4	69.6	3.3	2.1	24.6	0.5
EDS Spot 5	38.1	2.4	22.5	36.0	1.0
EDS Spot 6	11.6	0.8	87.0	0.4	0.2

b)



Points	O (at %)	Na (at %)	Al (at %)	Si (at %)	K (at %)
EDS Spot 1	58.9	2.3	1.6	36.5	0.8
EDS Spot 2	12.7	0.7	76.3	10.1	0.3
EDS Spot 3	18.8	1.8	8.3	69.4	1.7
EDS Spot 4	55.9	2.7	17.2	23.6	0.6
EDS Spot 5	29.3	2.9	22.2	40.5	5.1
EDS Spot 6	55.6	2.2	1.8	39.8	0.6

Fig. 15. EDS point analysis of the bonding interface for Sample C (Blue zone).

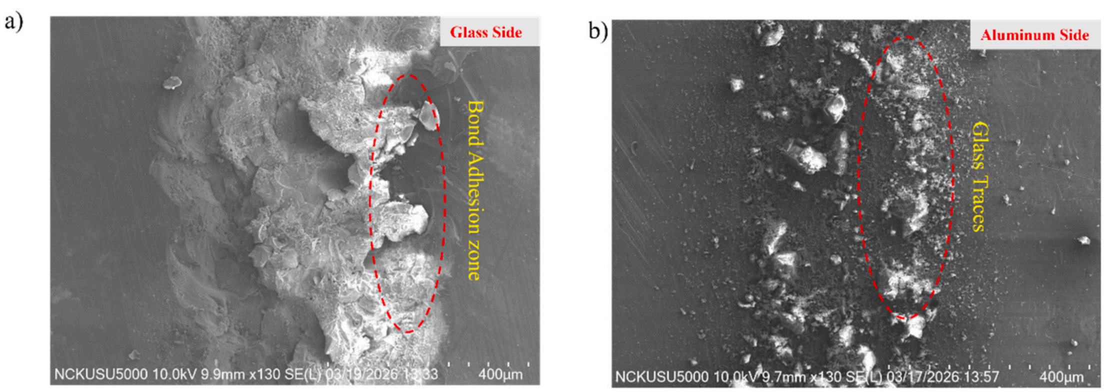


Fig. 16. SEM image of fracture surface morphology (a) Glass side, (b) Aluminum side (Green Zone).

detected in the reaction zones (e.g., Spot 3 at 87.7 at% Al), indicating that metallic Al has successfully infiltrated the glass surface. The consistent presence of Na and K across multiple spots suggests the formation of a complex, continuous Al–O–Si reaction layer. This multi-elemental transition zone indicates that the laser-induced melt pool enabled a deep chemical reaction rather than simple mechanical contact, which is essential for high-strength hermetic sealing.

In contrast to the Green Zone, the Blue Zone specimen (Fig. 15) exhibits markedly different, less integrated interfacial chemistry. While glass-derived elements are still present, their concentrations on the Aluminum side (Fig. 15(a)) are generally lower or more inconsistent compared to the Green Zone. For example, Spot 1 and Spot 6 on the aluminum remain almost entirely metallic (>87 at% Al), suggesting that large areas of the interface did not undergo significant chemical fusion. On the Glass side (Fig. 15(b)), while some Al migration is observed (e.g., Spot 2 at 76.3 at% Al), the Si levels remain high in many areas, indicating a more fragmented reaction layer.

Taken together, the EDS data suggest that the Green Zone produces a superior bond. The laser parameters enable the formation of a stable, hybridized Al–O–Si network spanning the interface. In the Blue Zone, the reaction is more superficial. The interface is dominated by metallic Al or by simple Al_2O_3 in contact with the glass, with limited penetration into the Si–O network. This explains the inferior mechanical performance of Blue Zone samples. The lack of a deep, continuous chemical transition layer reduces the joint's effective load-transfer capability compared to the optimized Green Zone conditions. The extreme brittleness of BF33 glass causes a high risk of interfacial 'pull-out' during polishing, making traditional cross-sectional measurement infeasible. Instead, thickness is qualitatively estimated by analyzing elemental transitions on fracture surfaces. The EDS interaction volume at 15–20 kV extends to 1.5-2.0 µm. In the Green Zone, high concentrations of glass-derived elements (26.3 at% Si, 66.2 at% O) remain on the Aluminum substrate, indicating a robust reaction layer 0.8-1.2 µm thick. In contrast, only trace Si is detected in the Blue Zone, suggesting a much thinner interface, less than 0.4 µm. This four-fold difference in reaction volume directly correlates to the superior load-bearing capacity of Green Zone joints. The Red Zone specimens were excluded from elemental mapping due to excessive thermal accumulation, which resulted in a brittle, over-melted interface. Since this regime was characterized by immediate thermal-stress-induced failure, the study focused on the optimized Green Zone where a stable Al–*O*–Si reaction layer was successfully maintained.

To further clarify the bonding mechanism and the resulting shear strength, SEM fractography was performed on the sheared surfaces of both regimes. As shown in Fig. 17 (Blue Zone), the fracture surface mainly shows interfacial failure with large regions of smooth glass and limited chemical interaction, as supported by the supplementary EDS analysis in Fig. S3 and S4. Some glass traces are visible on the aluminum side. Large regions of the glass surface are smooth, with localized cracking. This suggests that bonding was mainly due to surface adhesion

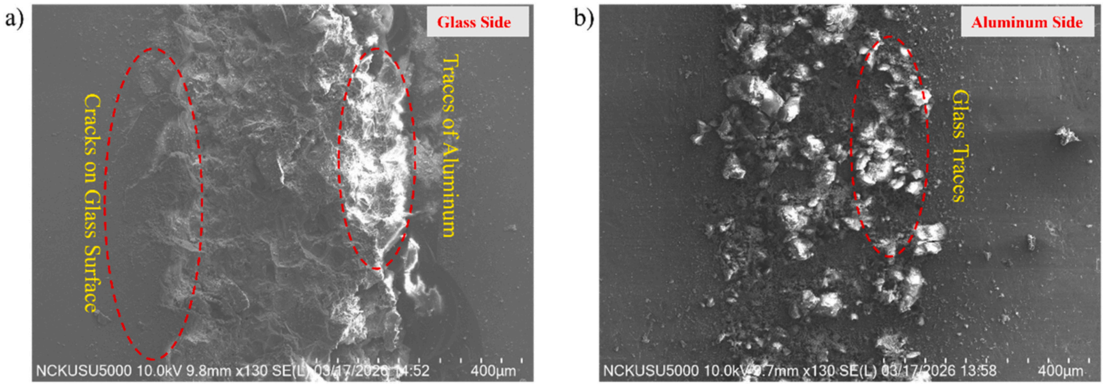


Fig. 17. SEM image of fracture surface morphology (a) Glass side, (b) Aluminum side (Blue Zone).

and limited chemical interaction, which explains the lower shear strength in this regime. In contrast, Fig. 16 (Green Zone) exhibits a clear cohesive failure mode within the BF33 glass. The aluminum side is heavily covered with glass fragments, while the glass side shows distinct 'traces of aluminum' and significant material pull-out. This indicates that the interfacial Al-O-Si chemical bond is stronger than the glass's intrinsic fracture toughness. The elemental mapping provided in the supplementary files (Fig. S1 and S2) confirms that the optimized energy input in the Green Zone facilitates a robust, high-integrity chemical fusion layer, forcing the fracture to propagate through the glass bulk rather than the joint interface.

6. Conclusions

In this study, an integrated numerical–experimental framework was developed to optimize the direct laser welding of BOROFLOAT® 33 glass to AA1050 aluminum using a wobbling nanosecond fiber laser. The research concludes the following.

1. **Multiphysics Modelling and Damage Mechanics:** A three-dimensional transient heat transfer model coupled with a phase-field fracture formulation successfully quantified the relationship between laser energy input and structural integrity. The simulations revealed that while high power and low scanning speeds increase the damage-driving energy density—promoting macroscopic cracking moderate, "optimized" energy input successfully confines thermal gradients. This confinement minimizes damage accumulation and significantly reduces the fracture risk at the brittle glass–metal interface.
2. **ANN-Driven Predictive Processing Map:** By training an ANN on the phase-field outputs, a high-resolution processing map was established for the power–speed design space. This map categorizes the welding outcomes into three physical regimes based on the damage variable (d):
   - **The Green Zone ($d < 0.3$):** Characterized by low predicted damage and high structural integrity.
   - **The Blue Zone ($0.3 < d < 0.85$):** Associated with internal porosity without macroscopic failure.
   - **The Red Zone ($d \geq 0.85$):** Defined by severe interfacial fracture and thermal damage.
3. **Validation of the Optimal Process Window:** Experimental trials verified the numerical predictions, identifying an optimal process window of 30–50 W and 3–5 mm/s within the "Green Zone." Joints produced in this regime exhibited crack-free interfaces and reached a peak shear strength of 8.5 MPa. This performance surpasses results typically reported for millisecond-laser glass-metal joints and approaches the quality of femtosecond laser welding, while offering significantly higher processing efficiency and lower power consumption.
4. **Interfacial Bonding Mechanisms:** Microstructural characterization (SEM/EDS) confirmed that superior mechanical performance in the Green Zone is driven by the formation of a continuous Al–O–Si reaction layer coupled with effective glass infiltration into the aluminium substrate. Conversely, the Blue Zone results in an Al/Al_2O_3 -dominated interface with limited penetration, highlighting that precise thermal control is mandatory for achieving chemical bonding in glass-metal systems.
5. **Broader Impact and Applications:** The proposed wobbling nanosecond-laser strategy provides a robust, interlayer-free solution for direct BF33–Al bonding. The methodology and the ANN-phase-field processing map serve as a quantitative toolset that can be readily adapted to other glass–metal combinations. This approach provides a reliable foundation for designing laser-welding processes in the optoelectronic, microfluidic, and advanced packaging industries.

Beyond BF33–Al bonding, the integrated phase-field and ANN framework offers a scalable approach for evaluating joinability in various dissimilar material pairs. By updating thermo-physical properties like CTE and fracture energy, the phase-field model can predict damage in other brittle-ductile systems such as high-strength aluminum alloys or soda-lime glasses. The ANN processing map is independent of the energy source, allowing recalibration for different lasers by retraining with material-specific thermal data. This methodology enables high-throughput virtual process prototyping, greatly reducing the experimental trial-and-error needed to establish stable welding windows for new optoelectronic and micromechanical applications.

Future research will build a statistical database to validate process reproducibility across specimens. The phase-field model for fracture evolution will be refined to include temperature-dependent and anisotropic glass properties. The ANN processing map will expand to cover more materials, like soda-lime glass and stainless steel, highlighting the versatility of the nanosecond-wobbling approach for advanced manufacturing.

AI Disclosure statement

During the preparation of this work, ChatGPT was utilized to refine the writing style, improve clarity, and ensure proper grammar and spelling. The AI tool was not used to generate or analyze any data presented in this work. The authors subsequently reviewed and edited the content to align with the intended academic rigor and take full responsibility for the final content of this publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence

the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jmrt.2026.03.272.

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